

MEICHENG XIONG

Minneapolis, MN, USA — xion2613@umn.edu

EDUCATION

University of Minnesota , Minneapolis, MN, USA	Sep 2023 - Present
— PhD student	GPA: 3.76/4.00
— Major: Geography	Supervisor: Di Zhu
Institute of Geographic Sciences and Natural Resources Research, CAS , Beijing, China	Sep 2020 - July 2023
— Master of Science	GPA: 3.88/4.00
— Major: Transport geography	Supervisor: Jie Huang, Jiaoe Wang (co-supervisor)
— Thesis title: <i>Investigating Spatial-Temporal Patterns of Elderly Travel Behavior: A Case Study of Kunming.</i>	
East China Normal University , Shanghai, China	Sep 2016 - June 2020
— Bachelor of Science	GPA: 83.1/100
— Major: Geographic sciences	

POSITIONS AND EMPLOYMENTS

Research Assistant: Rural Mobility and Access: Leveraging Big Data Analytics and Context-Aware Computing	Minneapolis, USA
<i>Center for Transportation Studies, University of Minnesota</i>	2024 Summer, 2025 Spring
Teaching Assistant: GEOG 3531/5531 Principles of Spatial Analysis	Minneapolis, USA
<i>Department of Geography, Environment and Society, University of Minnesota Twin Cities</i>	2025 Fall - Present
Teaching Assistant: GEOG 1502 Mapping Our World	Minneapolis, USA
<i>Department of Geography, Environment and Society, University of Minnesota Twin Cities</i>	2023 Fall - Present

SCHOLARSHIPS AND AWARDS

Student Honors Paper Competition (Finalist)	San Francisco, USA
<i>AAG Geographic Information Science and Systems Specialty Group</i>	2026
2026 Research Travel Funding	Minneapolis, USA
<i>Department of Geography, Environment and Society, University of Minnesota Twin Cities</i>	2025 Fall - 2026 Spring
2025 Research Travel Funding	Minneapolis, USA
<i>Department of Geography, Environment and Society, University of Minnesota Twin Cities</i>	2024 Fall - 2025 Spring
Student Scholar Award in the Graduate Category	Duluth, USA
<i>Minnesota GIS/LIS Consortium</i>	2024
Academic Scholarship for Graduate Students	Beijing, China
<i>Institute of Geographical Sciences and Natural Resources Research, CAS</i>	2021 Fall - 2022 Spring
Model Student of Academic Records	Shanghai, China
<i>School of Geographic Sciences, East China Normal University</i>	2018 Fall - 2019 Spring
Academic Scholarship for Undergraduate Students	Shanghai, China
<i>School of Geographic Sciences, East China Normal University</i>	2018 Fall - 2019 Spring
Successful Participant of Interdisciplinary Contest in Modeling	Shanghai, China
<i>Consortium for Mathematics and its Applications, USA</i>	2019
Academic Scholarship for Undergraduate Students	Shanghai, China
<i>School of Geographic Sciences, East China Normal University</i>	2017 Fall - 2018 Spring
Academic Scholarship for Undergraduate Students	Shanghai, China
<i>School of Geographic Sciences, East China Normal University</i>	2016 Fall - 2017 Spring

PUBLICATIONS

- Xiong, M., Zhu, D. (2025). Sensing lakeplace: Interactions between lakes and human activities. *Environment and Planning B: Urban Analytics and City Science*, 0(0), 1-20. DOI: [\[10.1177/23998083251386148\]](https://doi.org/10.1177/23998083251386148)

- **Xiong, M.**, Zhu, D. (2025). A Graph-based Deep Population Downscaling Model on Irregular Spatial Units. *In Proceedings of the 8th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, 2025, November, 1008-1018. DOI: [\[10.1145/3764912.3770815\]](https://doi.org/10.1145/3764912.3770815)
- Song, Y., **Xiong, M.**, Zeng, X., Zhu, D. (2025). A Behavior-Informed and Geo-Context-Aware Home Detection Framework for Mobile Phone Positioning Data. *In Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems*, 2025, November, 1008-1018. DOI: [\[10.1145/3748636.3764171\]](https://doi.org/10.1145/3748636.3764171)
- **Xiong, M.**, Zhu, D., Van Riper, D. (2025). A visitor-enriched census in the US cities using large-scale mobile positioning data. *Scientific Data*, 12(1), 1106. DOI: [\[10.1038/s41597-025-05410-0\]](https://doi.org/10.1038/s41597-025-05410-0)
- Huang, J., **Xiong, M.**, Wang, J. (2025) Understanding urban spatial structure through the lens of multiple modal accessibility. *Travel Behaviour and Society*, 40, 101043. DOI: [\[10.1016/j.tbs.2025.101043\]](https://doi.org/10.1016/j.tbs.2025.101043)
- Xiao, F., Mo, H., Wang, J., Dobruszkes, F., **Xiong, M.** (2025) Passenger winners and losers in the multi-airport system era: Evidence from Beijing tracking individual travel activities. *Transportation Research Part A: Policy and Practice*, 199, 104596. DOI: [\[10.1016/j.tra.2025.104596\]](https://doi.org/10.1016/j.tra.2025.104596)
- Huang, J., **Xiong, M.**, Wang, J., Cheng, L., Yang, H. (2024) Investigating socio-spatial differentiation for metro travelers using smart card data: Older people vs. others. *Applied Geography*, 165: 103230. DOI: [\[10.1016/j.apgeog.2024.103230\]](https://doi.org/10.1016/j.apgeog.2024.103230)
- Xiao, F., Wang, J., **Xiong, M.**, Mo, H. (2024) Does spatiotemporal heterogeneity matter? Air transport and the rise of high-tech industry in China. *Applied Geography*, 162: 103148. DOI: [\[10.1016/j.apgeog.2023.103148\]](https://doi.org/10.1016/j.apgeog.2023.103148)
- Huang, J., **Xiong, M.**, Wang, J. (2022) Route choice and parallel routes in subway networks: A comparative analysis of Beijing and Shanghai. *Tunnelling and Underground Space Technology*, 128: 104635. DOI: [\[10.1016/j.tust.2022.104635\]](https://doi.org/10.1016/j.tust.2022.104635)
- **Xiong, M.**, Huang, J., Wang, J., Yang, H. (2022) Spatial-temporal characteristics of the elderly subway travel based on activity space: A case study of Kunming. *Tropical Geography*, 42(12):2052-2062. (in Chinese)
- Wang, J., **Xiong, M.**, Huang, J. (2022) Subway accessibility under space-time constraints: A case study of Beijing. *Scientia Geographica Sinica*, 42(01):83-94. (in Chinese)
- Xiao, F., Mo, H., Wang, J., **Xiong, M.** (2023) Concept and identification of multiple airport systems: A geographical empirical research from a geographical perspective. *Scientia Geographica Sinica*, 43(04):606-616 (in Chinese)
- Zhao, T., Li, Z., Song, L., **Xiong, M.**, Liao, Y., Pei, T. (2022) Research on the relationship between floating population's sentiments and distribution of working and living in Beijing based on microblog data. *Journal of Geo-information Science*, 1-13[2022-09-02]. (in Chinese)

PRESENTATIONS

- **Xiong, M.**, Zhu, D. A Visitor-Enriched Census in U.S. Cities Using Large-Scale Mobile Positioning Data. *Geography, Environment & Society's Brown Day Celebration*, Minneapolis, USA, Apr 2026.
- **Xiong, M.**, Zhu, D. A graph-based deep population downscaling model on irregular spatial units. *2026 American Association of Geographers (AAG) Annual Meeting*, San Francisco, USA, Mar 2026.
- **Xiong, M.**, Zhu, D. A graph-based deep population downscaling model on irregular spatial units. *The 8th ACM SIGSPATIAL International Workshop: GeoAI '25*, Minneapolis, USA, Nov 2025.
- Song, Y., **Xiong, M.**, Zeng, X., Zhu, D. A behavior-informed and geo-context-aware home detection framework for mobile phone positioning data. *The 33rd ACM International Conference on Advances in GIS*, Minneapolis, USA, Nov 2025.
- Zeng, X., Song, Y., Zhu, D., **Xiong, M.** A behavior-informed and geo-context-aware home detection framework for mobile phone positioning data. *2026 CTS Transportation Research Conference*, Minneapolis, USA, Nov 2025.
- **Xiong, M.**, Zhu, D. A Visitor-Enriched Census in U.S. Cities Using Large-Scale Mobile Positioning Data. *2025 Annual Meeting of the Association of Collegiate Schools of Planning (ACSP)*, Minneapolis, USA, Oct 2025.
- **Xiong, M.**, Zhu, D. A Visitor-Enriched Census in U.S. Cities Using Large-Scale Mobile Positioning Data. *Minnesota GIS/LIS Consortium Annual Conference*, Minneapolis, USA, Oct 2025.
- **Xiong, M.**, Zhu, D. A Visitor-Enriched Census in U.S. Cities Using Large-Scale Mobile Positioning Data. *2025 Research Computing Exhibition by Minnesota Supercomputing Institute*, Minneapolis, USA, Apr 2025.
- **Xiong, M.**, Zhu, D. Lakeplace: Sensing interactions between lakes and human activities. *Geography, Environment & Society's Brown Day Celebration*, Minneapolis, USA, Oct 2024.
- **Xiong, M.**, Zhu, D. Lakeplace: Sensing interactions between lakes and human activities. *The 31st International Conference on Geoinformatics*, Toronto, Canada, Aug 2024.
- Huang, J., **Xiong, M.**, Wang, J. Understanding urban spatial structure through the lens of multiple modal accessibility. *2024 World Symposium on Transport and Land Use Research*, Bogota, Colombia, Jun 2024.
- **Xiong, M.**, Zhu, D. Lakeplace: Sensing interactions between lakes and human activities. *CaGIS and UCGIS Symposium 2024*, Columbus, USA, Jun 2024.
- **Xiong, M.**, Zhu, D. Lakeplace: Sensing interactions between lakes and human activities. *2024 American Association of Geographers (AAG) Annual Meeting*, Honolulu, USA, Apr 2024.
- Xiao, F., Mo, H., Wang, J., Dobruszkes, F., **Xiong, M.** Passenger winners and losers in the multi-airport system era: Evidence from Beijing tracking individual travel activities. *9th EUGEO Congress*, Barcelona, Spain, Sep 2023.
- **Xiong, M.**, Huang, J., Wang, J., Yang, H. Spatial-temporal characteristics of the elderly subway travel based on activity space: A case study of Kunming. *Anniversary Seminar of Economic Geography Committee of China*, Online, Oct 2021.
- **Xiong, M.**, Huang, J. *Progress of urban boundary and spatial structure identification based on big data*. China Conference on Geography, Beijing, China, Nov 2019. Oral presentation.

COURSEWORKS

Course project: Enhancing areal interpolation with graph-based modeling for spatial disaggregation of socio-demographic attributes

CEGE 8490 Generative AI in Transportation Research 2025 Fall

Course instructors: Dr. Seongjin Choi

Course project: Activity sequence generation using classical deep generative models: A comparative analysis

GEOG 5900 Geospatial + X 2024 Fall

Course instructors: Dr. Eric Shook

Course project (collaborative): Mapping our Westbank: utilizing 3D modeling techniques

GEOG 5533 Advanced Spatial Analysis 2024 Spring

Course instructors: Dr. Di Zhu

Course project: Lakeplace: Sensing interactions between lakes and human activities

CSCI 5523 (001) Introduction to Data Mining 2024 Spring

Course instructors: Dr. Vipin Kumar, Dr. Michael Steinbach

GEOG 8001 Geographic Thought 2023 Fall

Course instructors: Dr. Arun Saldanha

Course project: Lakeplace: Sensing interactions between lakes and human activities

GEOG 8294 Spatiotemporal Modeling and Simulation 2023 Fall

Course instructors: Dr. Ying Song

Course project: Residential area detection based on PlaceIQ dataset: A case study in the TCMA

Applications of Geographical Information System 2021 Spring

Course instructors: Dr. Fenzhen Su, Dr. Yunyan Du, Dr. Quanqin Shao

Spatial Data Mining 2021 Spring

Course instructors: Dr. Tao Pei

Course project: Research on the relationship between the floating population's sentiments and distribution of working and living in Beijing based on microblog data

Spatial Analysis 2020 Fall

Course instructors: Dr. Tao Pei, Dr. Ting Ma, Dr. Yunyan Du, Dr. Ci Song

Digital Map Making and Using 2020 Fall

Course instructors: Dr. Qingwen Qi

Industry and Transport Geography 2020 Fall

Course instructors: Dr. Jiaoe Wang, Dr. Xiaoping Zhang

Course project: Subway accessibility under space-time constraints: A case study of Beijing.

RESEARCH INTERESTS

- Social sensing
- Geospatial Artificial Intelligence
- Geographic information science and big geo-data
- Human mobility

LANGUAGE

- Chinese [native]
- English [fluent]
- German [beginner]
- French [beginner]

SKILLS

- **Programming Language** –Python, R, SQL
- **Software** –ArcGIS, MATLAB, Stata, SPSS, HTML, LaTeX, Adobe Photoshop, Microsoft Office
- **Skill** –Spatial analysis, geovisualization, big data mining, statistical analysis, machine learning, deep learning